

SHROFF

The MSME Financial Health Card: a digital shroff for the credit-invisible.

Alternate-data credit decisioning: reads a business's consented GST · UPI/bank (AA) · EPFO money-trail, scores it 300–900 with signed reasons, screens ~45,000 real government registry rows for phoenix fraud, and emits an OCEN 4.0 loan offer.

Team name	SHROFF
Team leader	Anshuman Atrey
Problem statement	Problem Statement 3: Financial Health Score

Brief about the idea

SHROFF · TRACK 03

Millions of Indian MSMEs (kirana stores, small traders, tiny manufacturers) are **new-to-credit**: they run on cash and UPI and never kept formal books. Banks judge them on audited statements, ITR and a CIBIL score they simply don't have, so viable businesses are auto-rejected. Meanwhile, some borrowers who *look* clean on paper are quietly failing, and get through.

THE IDEA: TELL THE TRUTH IN BOTH DIRECTIONS

- **Approve the invisible-but-healthy.** Read the business's real, **consented** digital money-trail (GST returns, UPI/bank flows via Account Aggregator, EPFO headcount) and score what the bureau can't see.
- **Catch the looks-fine-but-failing.** Compare GST-declared turnover against bank-verified inflows on the same ₹ axis; flag bounce streaks, sliding receipts, distressed buyers.
- **Catch the phoenix.** A spotless new company whose promoter's network reaches a struck-off / wilful-defaulter shell → deterministic REFER, regardless of score.

WHAT THE BANK RECEIVES

- A **300–900 Financial Health Score** with four sub-scores: cash-flow · growth · stability · compliance.
- **Signed reason codes** (TreeSHAP): every point on the score has an auditable reason. **No LLM ever computes a number.**
- A verdict, **APPROVE** **REFER** **DECLINE**, with amount & tenure, emitted as an **OCEN 4.0-aligned loan offer** with risk-based pricing.

Why "Shroff"? The traditional Indian banker who judged merchants by cash-flow and conduct, never paperwork. This is that instinct, rebuilt from alternate data.

Opportunities & USP

HOW IT DIFFERS · HOW IT SOLVES

HOW IS IT DIFFERENT FROM EXISTING IDEAS?

- **Two-sided decisioning.** Most inclusion tools only widen approval. SHROFF also tightens it: the same data catches clean-on-paper decliners, so the book gets **healthier, not just larger**.
- **Deterministic core, generative shell.** Monotonic-constrained LightGBM + TreeSHAP. Any LLM use is env-gated narrative only, regulator-defensible by construction.
- **A third signal nobody scores: the entity graph.** PAN-spine graph walk over real registries detects **phoenix fraud**: the promoter's network, not just the applicant.
- **Real registry data in the PoC.** ~45,000 genuine government rows (MahaGST, SEBI/NSE, CBDT, MCA, RBI, OpenSanctions), not mocked.

HOW IT SOLVES THE TRACK 03 PROBLEM

- High NTC/NTB rejection → scores the consented alternate-data trail the statement can't show.
- No unified framework → one health card: score + sub-scores + screening + EWS + offer.
- Slow onboarding → near-real-time API (<100 ms/score), AA-consent native, ULI/OCEN adapter-ready.

USP IN ONE LINE

"Sees the invisible borrower, in both directions: approves the invisible-but-healthy, catches the looks-fine-but-failing, and refers the phoenix the score alone would have lent to."

List of features offered by the solution

ALL WORKING IN THE MVP TODAY

1 · SCORING & EXPLAINABILITY

- **300–900 health score** from ~45 engineered signals; 4 sub-scores (cash-flow · growth · stability · compliance) via monotonic LightGBM ensemble.
- **Calibrated 12-month PD** (isotonic) behind every score.
- **Signed TreeSHAP reason codes:** auditable, per-applicant.
- **What-if simulator:** how the score moves if inputs change.
- **Early-warning strip:** bounces, headcount drops, GST late/nil streaks.

2 · SCREENING & FRAUD GRAPH

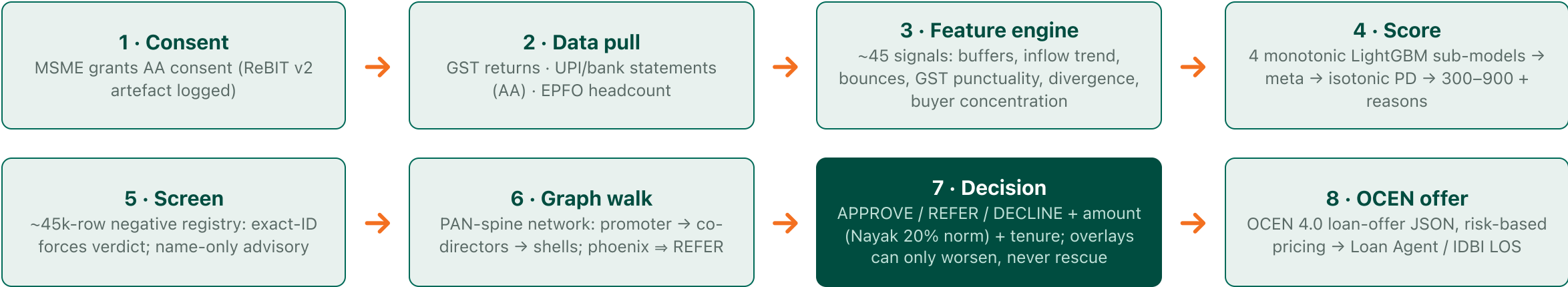
- **Negative-registry screening** over ~45,000 real government rows: exact-ID hits (PAN/GSTIN/CIN/DIN) force the verdict; name-only hits stay advisory (bank AML pattern).
- **PAN-spine entity graph:** walks promoters, co-directors, shared addresses; **phoenix detection** auto-REFERS.
- **GST-vs-bank divergence view:** declared vs verified turnover, same ₹ axis.

3 · LENDING RAIL & CONSOLE

- **OCEN 4.0-aligned loan offer:** risk-based pricing (Band A 13.77% → C 18.14%), reducing-balance EMI, tenure, validity.
- **AA consent artefacts** (ReBIT v2 shape): consented, not scraped.
- **Underwriter console:** portfolio ledger, health card, verdict stamp, entity graph.
- **Fixture fallback:** console degrades gracefully if the API is down; a chip shows LIVE vs FIXTURE.

Process flow

CONSENT → SCORE → SCREEN → DECIDE → OFFER



THREE DEMO PERSONAS PROVE THE FLOW END-TO-END

Persona	Story	Score	Verdict
Ramesh Kirana	No CIBIL; 24 months clean bank + GST cash-flow	769 • A	APPROVE ₹12.5L @ 13.77%
Suresh Trading	GST declared ~37% above bank inflows; bounces, sliding receipts	419 • E	DECLINE
Nexon Trading	Tidy thin file; promoter's network reaches a struck-off shell	678 • C	REFER • phoenix graph

Wireframes: the Underwriter's Ledger

NOT MOCK-UPS: THE RUNNING PRODUCT

DRISHTI

2021 INNOVATE 2026 · TRACK 03

OPEN CONSOLE →

THE MSME FINANCIAL HEALTH CARD

Sees the invisible borrower —
in both directions.

Six crore MSMEs run real businesses that credit bureaus cannot see; others look clean on paper while their cash flow quietly burns. DRISHTI reads the consented ledger — bank, GST, EPFO — scores it with an explainable model, and screens every identifier against the public negative registries. One card. Both failure modes.

OPEN THE UNDERWRITER'S CONSOLE →

VIEW A LIVE HEALTH CARD

6.3 Cr

₹25L Cr

24 mo

< 60 s

registered MSMEs in India

estimated formal credit gap

consented cash-flow window

scores, reasons & sanction

HOW THE CARD IS WRITTEN

01

Cash-flow truth, consented

24 months of bank inflows, GST filings and EPFO payrolls fetched over Account Aggregator calls under a purpose-bound ReBET consent. The borrower's real ledger — not a credit bureau's silence.

02

Deterministic score, signed reasons

Four monotonic gradient-boosted sub-models — cash flow, growth, stability, compliance — calibrated to a 12-month PD, each decision carrying TreeSHAP reason codes. Nothing generative ever computes a number.

03

Registry screening & the entity graph

Every identifier walked through MCA stack-off, GST non-genuine, willful-defaulter and DIN registries — then along the promoter graph, where phoenix operators hide behind fresh CINs at old addresses.

THE BOTH-DIRECTIONS THESIS

Direction one — the deserving invisible

A kirana store with no credit history but two years of clean, growing, GST-consistent cash flow deserves working capital. The onboarding head prices it: score, band, sanction amount under the 20% working-capital norm, tenure — with every reason signed.

Direction two — the clean-on-paper risk

Declared turnover 40% above bank receipts. Bounces stacking up. A promoter whose last company was struck off in May and reborn in June at the same address. The early-warning head and the entity graph catch what a bureau score never will.

THREE BORROWERS, THREE VERDICTS

045650001

Ramesh Kirana & General Stores

APPROVE

No credit history, no collateral — but 24 months of

009550002

Suresh Trading Co.

DECLINE

Clean on paper. But declared GST turnover runs 40% above bank inflows, receipts are oldfile, and turnover

PH0051003

Nexon Trading Pvt Ltd

REFER

A tidy thin file the score alone would pass. The entity graph walks promoter DIN to a struck-off shell at the

Story page · the thesis: "sees the invisible borrower, in both directions"

'S CONSOLE

APPROVED

1

REFERRED

1

DECLINED

1

	SEGMENT	REQUESTED	SCORE	VERDICT
Stores	kirana_retail The invisible-but-healthy borrower: growing kirana st	₹15,00,000	769 band A	APPROVE
	trader Clean-on-paper-but-risky: inflows down ~20% in 6 m	₹10,00,000	419 band E	DECLINE
	trader The phoenix borrower: 14-month-old company, clean	₹20,00,000	678 band C	REFER

ic LightGBM ensemble with TreeSHAP reason codes, calibrated to a 12-month PD. Registry and phoenix overlays are deterministic and can only worsen a d data.

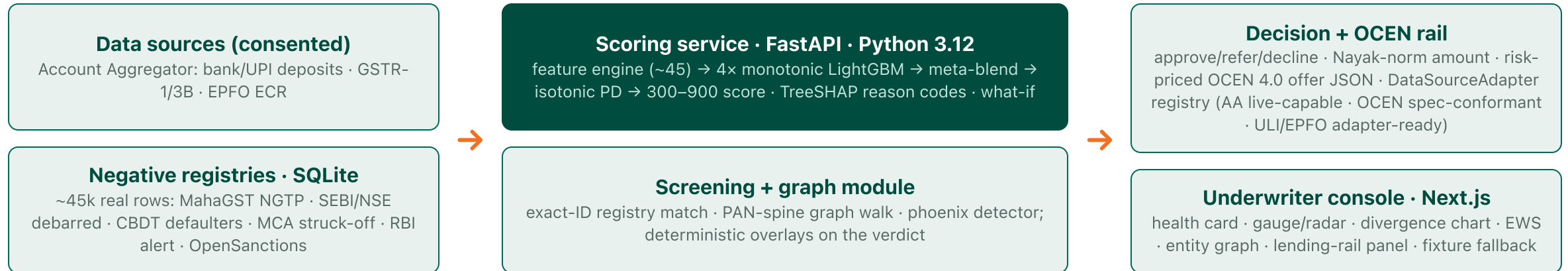
Portfolio console · all personas scored live; 1 approved / 1 referred / 1 declined

DESIGN LANGUAGE

- A **ledger, not a dashboard**: verdict stamp, rupee-axis charts, reason codes read like an underwriter's file.
- Gauge + radar for the 4 sub-scores; React Flow entity graph with flagged nodes in oxide red.
- **LIVE API / FIXTURE MODE** chip: honest about what's being served.
- Every registry row carries an **is_sample** chip: real vs honestly-labeled sample.

Architecture

DETERMINISTIC CORE · GENERATIVE SHELL



ARCHITECTURE PRINCIPLES

- **No LLM on the decision path:** generative is env-gated narrative only; nothing generative computes a number.
- **Overlays only worsen:** screening/graph can downgrade a verdict, never rescue one.
- **Deterministic + reproducible:** seed-42 pipeline, artifacts trained offline, loaded at boot, <100 ms/score.
- **Swappable data seam:** SQLite in the PoC; the adapter layer makes AA-live / Postgres a config change.

Technologies used

BORING WHERE IT SHOULD BE · SHARP WHERE IT COUNTS

Layer	Stack	Why
Web console	Next.js (App Router, TypeScript) · Tailwind · Recharts · React Flow	Instant deploy link (Vercel); React Flow renders the entity graph natively
Scoring API	Python 3.12 · FastAPI · LightGBM (monotonic) · SHAP · scikit-learn	Monotone constraints = regulator-friendly; TreeSHAP = exact reason codes; <100 ms/score
Screening + graph	Stdlib SQLite + PAN-spine graph walk (same service)	Multi-key relational lookups (PAN/GSTIN/CIN/DIN joins); deterministic, zero extra infra
Data (PoC)	Synthetic parquet (8,000 MSMEs, seed 42) + ~45k real registry rows in-repo	Judge-proof: no network, no accounts, cold-clone reproducible
Rails	AA (ReBIT v2 consent artefacts) · OCEN 4.0 offer schema · ULI adapter seam	Track 03 asks for ULI/OCEN/AA integration; honest status per adapter
Hosting	Vercel (web) · Render (API), free tier	Live deployment link for judging: shroff.onrender.com

PRODUCTION PATH (POST-SHORTLIST)

Same pipeline, different plugs: Supabase Postgres for consent logs & portfolio persistence · AA FIU-module live connection · retrain + recalibrate on IDBI sandbox portfolio outcomes · optional LLM narrative stays env-gated at the edge.

Estimated implementation cost

POC → PILOT → PRODUCTION

₹0

PoC as demoed: free tiers (Vercel + Render), in-repo data, no API keys, no licences

WHY THE POC IS ₹0

- Open-source stack end-to-end (MIT-licensed repo).
- Models train offline in ~40 s on a laptop; no GPU, no training cloud.
- Registry data captured from public government sources.

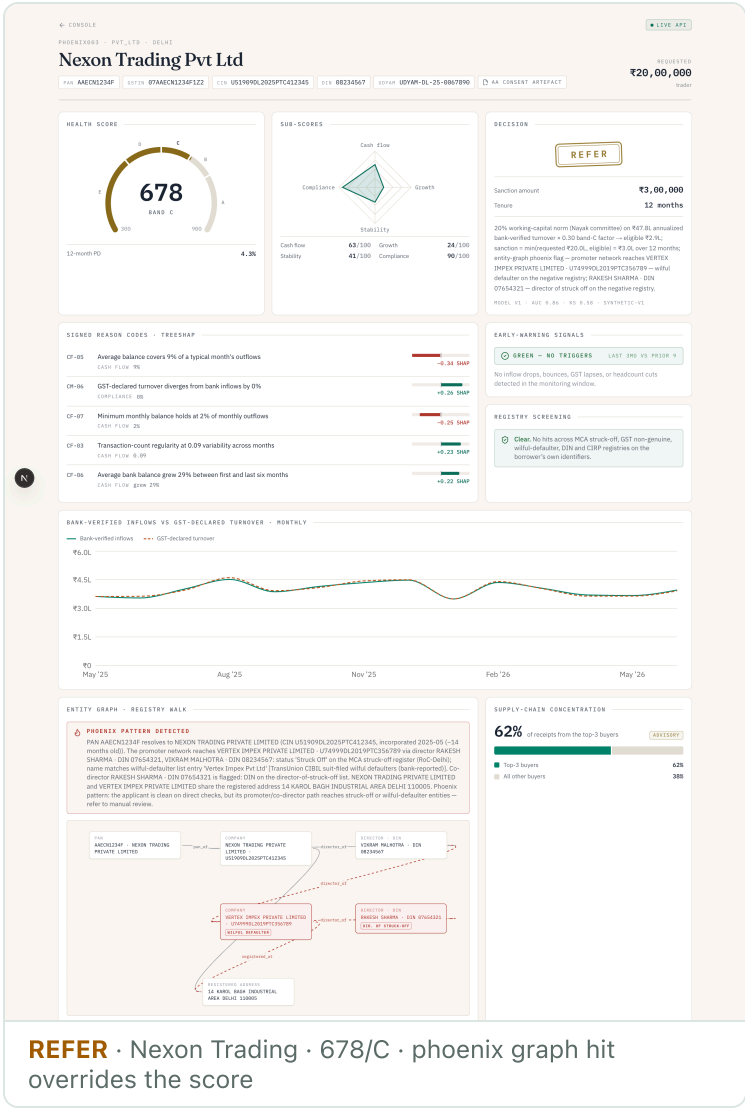
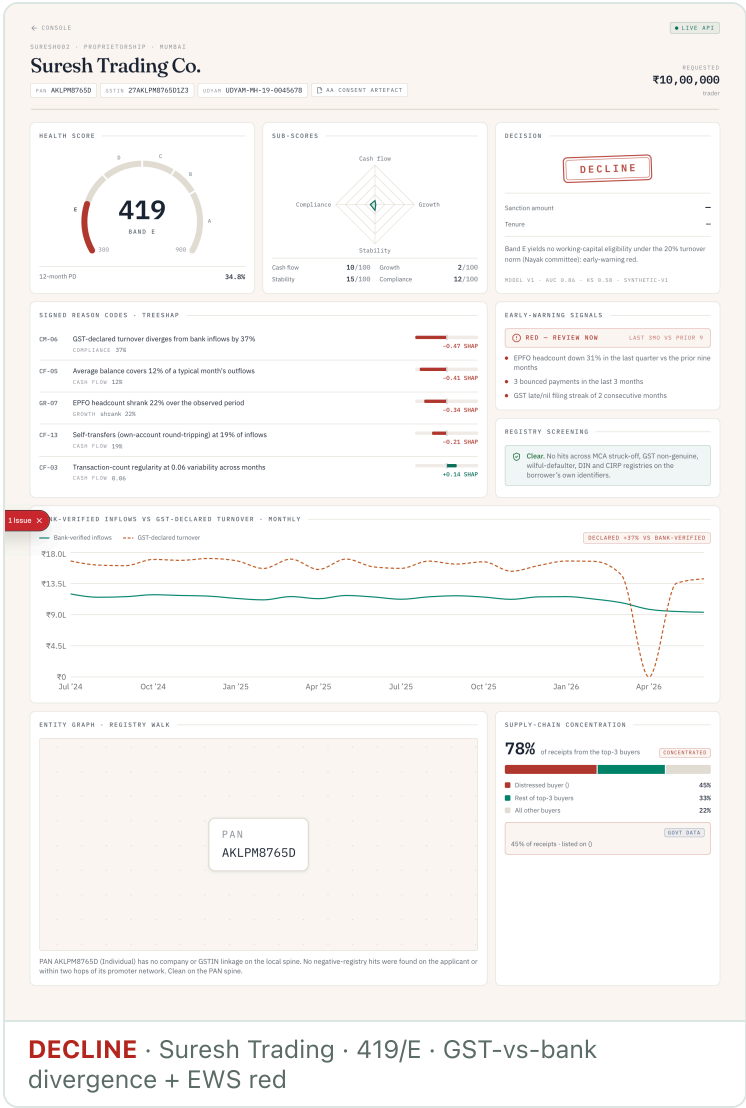
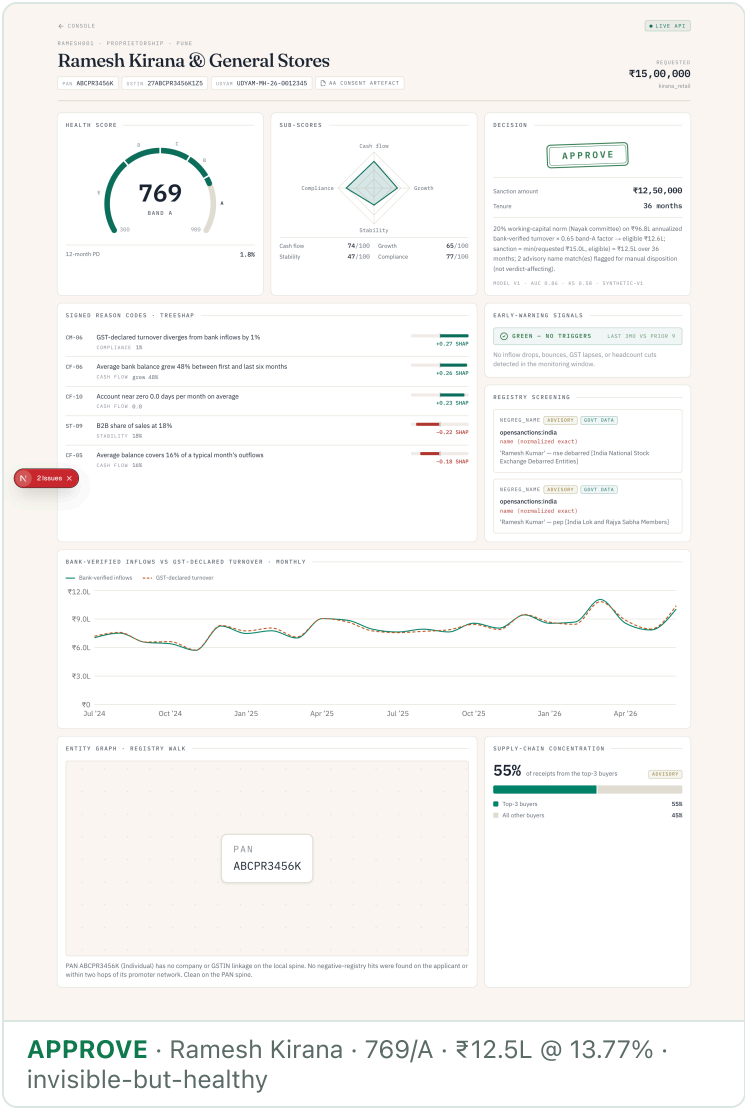
PILOT & PRODUCTION: INDICATIVE ANNUAL RUN-RATE

Item	Pilot (~5k assessments/yr)	Scale (~100k/yr)
Cloud infra (API + web + Postgres)	₹1–2 L	₹8–15 L
AA data-pull fees (per-consent, market ₹5–25)	₹0.5–1.5 L	₹10–25 L
Registry/data subscriptions (MCA bulk, watchlists)	₹1–3 L	₹3–6 L
MLOps: monitoring, retraining, model validation	₹2–4 L	₹6–12 L
Total (ex-manpower)	₹5–10 L	₹27–58 L

Per-decision cost at scale: ~₹30–60. Against the margin on a single healthy ₹12.5L working-capital loan, the system pays for itself within the first few hundred approvals it makes possible.

Snapshots of the prototype

THREE VERDICTS · ONE CONSOLE



Performance report & benchmarking

UNTOUCHED HOLDOUT · N=1,600 · REPRODUCIBLE (SEED 42)

0.856

AUC (KS 0.575), combined score on holdout;
LogReg regulator-view baseline 0.867 shipped
side-by-side

7.6×

cleaner approved book: 1.36% bad-rate in
approved A/B vs 10.24% population

94.5%

of eventual defaulters kept out of the approved
book at a 41.4% approval rate

MONOTONE RANK-ORDERING BY BAND (OBSERVED DEFAULT RATE)

Band	n	Score range	Default rate
A	637	751–900	1.26%
B	26	689	3.85%
C	436	660–678	4.36%
D	199	542–578	11.56%
E	302	300–483	37.42%

HONESTY BOX: WHAT THESE NUMBERS DO AND DON'T CLAIM

- Training set is **synthetic** (8,000 MSMEs encoding known risk economics). The numbers prove the **pipeline** (feature engine, monotone constraints, calibration, decisioning), not production performance.
- In production the same pipeline retrains and recalibrates on the bank's own portfolio outcomes (sandbox phase).
- Full reproducible proof page ships in-repo: [docs/pipeline-proof.html](#); metrics in [ml/artifacts/](#). Cold rebuild reproduces AUC/KS exactly.
- All 3 personas verified through the **live deployed API**: input → score → verdict → OCEN offer.

Future development

FROM HEALTH CARD TO LENDING PLATFORM

PROTOTYPE PHASE (IF SHORTLISTED · AUG 2–16)

- Swap synthetic data for **IDBI sandbox APIs/datasets**; retrain + recalibrate via the same seed-locked pipeline.
- AA FIU-module live connection (adapter seam already in place).
- Scale MCA struck-off registry from 900 to the full **129,694 rows** (registered data.gov.in key).
- Supabase Postgres for consent logs + portfolio persistence.

PLATFORM ROADMAP

- **Continuous monitoring**: re-score the live book monthly from refreshed AA/GST pulls; EWS alerts to the RM.
- **Dynamic credit limits**: working-capital line moves with the verified cash-flow, not a yearly review.
- **Supply-chain risk**: screen counterparties against the negative registry (distressed-buyer flags already prototyped).
- **Adverse-media overlay**: news-scanner adapter on the compliance sub-score.
- **Track 04 synergy**: same engine, default-prediction label: a 12-month early-warning score for the existing MSME book.

Improvements — Phase 2

REFINED PROTOTYPE · DEVELOPED SINCE SHORTLISTING

FROM 3 PERSONAS TO ANY REAL BUSINESS

- **Live GSTIN/PAN lookup:** score any real identifier, not just the 3 demo personas — deterministically simulated financials, seeded from the identifier itself, so the same input always reproduces the same score.
- **Live document upload/download:** download a sample `business_profile.csv` + `monthly_history.csv`, edit the numbers, upload them back — the score reflects exactly what's in the file, not a re-simulation.
- Registry + entity-graph checks still run on the real ~45,000-row government dataset regardless of path — a flagged real GSTIN still force-declines even on edited financials.

MODEL TRANSPARENCY & LENDING DEPTH

- New **Model & Score tab:** the exact scorecard formula, band table, per-sub-model validation metrics, and monotone-constraint counts, read live from the trained bundle — never hardcoded.
- **Population-rank percentile** on every score: "healthier than X% of N reference MSMEs," computed from the same trained model.
- **Second loan type:** DSCR-based Term Loan sizing alongside the existing Nayak-norm Working Capital line.
- **Bank operational-impact overlay:** quantifies onboarding + field-verification cost saved when an application auto-clears, with cited industry benchmarks, honestly zeroed whenever a human reviewer is still needed.

CONSOLE RESTRUCTURED INTO A REAL DASHBOARD

Persistent navigation shell (Portfolio · Live demo) replaces a single scrolling page; the health card itself is now tabbed — Overview · Model & Score · Cash flow / GST · Screening & Graph · Supply chain / Rail — so a reviewer finds any one signal in one click instead of scrolling the whole file.

GITHUB PUBLIC REPOSITORY

github.com/AnshumanAtrey/shroff

MIT-licensed · cold-clone runbook in RUN.md · reproducible pipeline (seed 42)

FINAL PRODUCT: LIVE DEPLOYMENT

shroff.vercel.app

Underwriter console · LIVE API chip when connected to the scoring service

SCORING API: LIVE

shroff.onrender.com/api/health

POST /api/score · GET /api/screen · /api/graph/{pan} · /api/ocen/offer/{id}

DEMO VIDEO (3 MINUTES)

loom.com/share/e0a3c715e5bf499a968a030774963232

The 3-persona walkthrough: Ramesh (approve) · Suresh (decline) · Nexon (phoenix refer)

TRY IT IN 30 SECONDS

```
curl -X POST https://shroff.onrender.com/api/score -H 'Content-Type: application/json' -d '{"msme_id":"PHOENIX003"}'
```

→ score 678 · band C · REFER, with the struck-off shell named in the rationale.

*"The shroff never asked a merchant for
paperwork.
He watched how the money moved."*

SHROFF: approve the invisible-but-healthy · catch the looks-fine-but-failing · refer the phoenix.

Thank you · Anshuman Atrey · IDBI Innovate 2026 · Track 03